

ASHWINI AHER

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SUMMARY

I am an Electrical and Telecommunications Engineer with a professional background spanning wireless communications, cellular systems, speech processing, and machine learning. I hold a Master of Science in Electrical Engineering from the State University of New York and a Master of Engineering in Electronics and Telecommunication from India. My career began with research in speech processing, including work on vocal tract shape estimation using Linear Predictive Coding, which resulted in a peer-reviewed IEEE conference publication in 2015. I subsequently gained industry experience in wireless communications and cellular modem technologies at Samsung Electronics America and Qualcomm, where I worked on 4G/5G modem integration, validation, system performance, NSA/SA scenarios, carrier aggregation, modulation, regression analysis, and technical debugging. More recently, I have been developing expertise in machine learning and applying it to wireless and speech-related problems, including speech-based channel quality classification and SINR prediction, while also building practical ML deployment skills using Python, Docker, AWS, Flask, and GitHub Actions/CI/CD. My long-term professional direction is to combine my background in wireless communications, speech/signal processing, and AI/ML to contribute to research and engineering in intelligent wireless communication systems, particularly emerging 5G/6G technologies.

WORK EXPERIENCE

Cellular System Test Engineer – Qualcomm, San Diego

July 2022-Dec 2023

- Used Qualcomm's Proprietary Tools and Automation framework to test software on latest modems for 4G/5G technologies.
- Executed 100+ 4G/5G test cases per release cycle to validate modem software using Qualcomm proprietary tools, covering SA/NSA call flow, uplink/downlink throughput for different 5G, 4G band combinations, Carrier Aggregation (CA), higher-order modulation schemes, different MIMO combinations.
- Worked on various testing equipment (TEs) like Keysight, Live Networks, Qualcomm in-house TE in lab environment.
- Evaluated performance KPIs (Key Performance Indicators) and monitored dashboards.
- Debugged test failures using QXDM, automation logs, and test setup analysis to identify root causes.
- Analyzed functional regressions, system crashes, stress tests and collaborated with development teams for timely resolution.
- Created CR (Change Request), JIRA ticket to track the issue.
- Created and maintained Confluence documentation for new findings, knowledge sharing and training.

Modem Integration and Test Engineer – Qualcomm, San Diego (Contract - Novus Resources, Inc.)

Nov 2020-July 2022

- Validated modem software across 5G NSA/SA and 4G use cases, identifying 30% more regressions compared to team average.
- Collaborated with cross-functional teams to integrate, validate and support commercialization of Qualcomm 5G products.

Software Quality Analyst – Samsung Electronics America, Overland Park (Contract - BTI Solutions, Inc.)

July 2020-Nov 2020

- Evaluated user interface software quality for SAMSUNG UE on field for Sprint/T-Mobile Carrier.
- Conducted different test cases related to 5G attach-detach, mobility, leg switch, EN-DC carrier aggregation, LTE only mode, throughput, different operators and their bands, different modes of tethering, MIMO, 64-QAM, 256-QAM, etc.
- Utilized diagnostic tools such as QXDM to collect logs, Umetrix Data for throughput and also collecting on-device logs.

Research Intern – University at Buffalo, NY, USA

Oct 2019-July 2020.

- Performed numerical tests and parameter tuning to produce results for 'Multi-Kernel Recursive Least Squares in the Presence of Sparse Outliers', using different combinations of Gaussian and Laplace Kernel.

Graduate Intern - FoodNerd, Buffalo, NY, USA

Feb 2019-May 2019

- Worked on Wearable Dietary Monetary Device (WDMD) project and created project proposal for WDMD project.

Graduate Student Assistant - University at Buffalo, NY, USA

May 2018-Dec 2018

- University Facilities - Finance and Administration Department.

Business Development Manager – Mighty Magpie, India

Jan 2017-July 2017

- Strategized advertising, marketing campaigns to promote products.

Assistant Professor - G. V. Acharya Institute of Engineering and Technology, India

June 2015-Aug 2015

- Taught the subjects, "Analog Electronics-1" to sophomores and "Optical Fiber Communication" to senior students.

Executive Engineer - Steam Airetric Controls, India

Sept 2010-July 2012

- Invented raw material management system for purchasing department and increased annual benefit almost by 97%.

- Led the production and purchasing departments and supervised effective working of them by scheduling weekly list of duties.
- Brought cost advantage equivalent to US\$10,000, by optimizing inhouse worker's workload instead of giving it to contractors.

TECHNICAL PROJECTS

- **Speech Based Channel Quality Classification (Python, PostgreSQL, Pandas, Scikit-learn, Librosa, VS Code)** April 2026
 - Developed a Speech-Based Channel Quality Classification System using the Google Speech Commands dataset.
 - Simulated channel conditions by adding AWGN noise at multiple SNR levels to generate realistic noisy speech signals.
 - Extracted audio features such as MFCC, delta MFCC, and spectral features using Librosa.
 - Performed data analysis and outlier detection in PostgreSQL, and created datasets with and without outliers.
 - Trained Logistic Regression and SVM models using Scikit-learn, achieving ~92% classification accuracy on test data.
- **Student Performance Indicator Project (Python, Flask, Docker, AWS EC2, GitHub Actions, CI/CD)** December 2025
 - Built an end-to-end ML pipeline for student performance prediction using Python, Pandas and Scikit-learn.
 - Containerized and deployed a Flask-based application on AWS EC2 instance using Docker to enable reproducible predictions.
 - Implemented a CI/CD pipeline using GitHub Actions to automatically build Docker images, push them to Amazon ECR, and deploy updated containers to EC2.
- **SINR Prediction Using Regression (Python, PostgreSQL, Pandas, Scikit-learn, VS Code)** March 2026
 - Built an end-to-end ML pipeline integrating PostgreSQL with Python for data ingestion, preprocessing, and modeling.
 - Simulated a wireless dataset (10,000 UEs) using link-budget principles, including features as distance, path loss, received power, and SNR.
 - Trained and evaluated regression models (Linear Regression, Ridge, Lasso) to predict SINR, achieving $R^2 \approx 0.93$ on test data.
- **Netflix Data Cleaning Data Analysis Project (PostgreSQL, Pandas)** January 2026
 - Performed data cleaning and transformation on Netflix dataset using SQL (CTEs, window functions).
 - Designed a normalized schema (ERD) and analyzed trends across director, genre, and country-wise content production.
- **Python SQL Data Analytics Project** Jul 2025 – Jul 2025
 - The objective of this project was to perform data analysis using SQL queries in PostgreSQL.
 - The required dataset is provided by Ankit Bansal and it was downloaded via Kaggle API.
 - Python was used to load and extract the dataset.
 - Pandas library was used for cleaning, manipulation and loading the dataset into PostgreSQL.
 - Finally, dataset was analysed using SQL queries in PostgreSQL database.
- **Simulated MIMO system to find SER (MATLAB)** May 2018 – May 2018
 - Implemented 2X1 and 2X2 MIMO systems using MATLAB. Applied Jake's fading simulator to generate Rayleigh fading channels for both the systems and generated Alamouti code.
 - Plotted Symbol Error Rate (SER) vs Signal to Noise Ratio (SNR) curve for both the systems. From the results, observed that 2X2 MIMO system performance is better than 2X1 MIMO system as diversity characteristics of MIMO improves system reliability.
- **Digit recognition using Support Vector Machines (Python, Scikit-learn)** March 2018- April 2018
 - Performed classification in Python on 'MNIST' dataset by applying SVM with linear kernel and Radial Basis Function (RBF).
 - The RBF kernel gave best results with test accuracy of 97.4% whereas linear kernel test accuracy was 93.78%
- **Classify diabetic patients using regression methods (Python, NumPy)** Feb 2018
 - Executed Linear Discriminant Analysis (LDA), Quadrature Discriminant Analysis (QDA) and Ridge regression techniques on 'diabetes' dataset using (Python, Matplotlib) and used Mean Squared Error (MSE) as performance matrix for comparison.
 - From the results, concluded that the Linear and ridge regression gave lowest MSE, as the data was linearly separable.
- **Entropy computation for a given English text (MATLAB)** Feb 2018
 - Implemented a code to calculate the entropy of a given English text, using the frequency of the letter. Also, computed the entropy for the different distributions.
- **Comparison of SER/BER of various modulation schemes for optimum receiver detection** Dec 2017
 - Simulated Binary-Pulse Amplitude Modulation (PAM), 4-Pulse Amplitude Modulation and 4-Quadrature Amplitude Modulation (QAM) and compared their results by plotting SER vs SNR and Bit Error Rate (BER) vs Signal to Noise Ratio (SNR) curves.
 - Observed that Binary-PAM gives the better SER, BER performance for a particular SNR followed by 4-QAM and 4-PAM.

PROFESSIONAL DEVELOPMENT PROJECTS

- **SEAS 360: Graduate Student Professional Development Program** August 2018 - May 2019
- Participated in "SEAS 360: Graduate Student Professional Development" program.
- **Navigate Project** January 2018 – May 2018
- Participated in NSF funded NAVIGATE Project for Professional Development.

EDUCATION

- **Master of Science in Electrical Engineering**, The State University of New York, USA GPA: 3.74 | Sept 2019
- **Master of Engineering in Electronics and Telecommunication**, Mumbai University, India GPA: 3.67 | Jan 2015
- **Bachelor of Technology in Electronics and Telecommunication**, Dr. B. A. T. U., India GPA: 3.6 | June 2010

SKILLS

- **Courses:** MIMO Wireless Communications, Principles of Modern Digital Communication, Information Theory and Coding, Big Data Analytics, Machine Learning, Deep Learning, Probability and Stochastic Process, Speech Processing
- **Concepts:** 5G/4G Architecture, NSA/SA/LTE Call Flow, Scheduling, Transmission Schemes, Air Interface, Protocol Stack, Carrier Aggregation, MIMO, OFDM, LTE, FDMA, TDMA, WCDMA, OFDMA, SC-FDMA, Modulation Techniques, Source Coding, Channel Coding
- **ML/DL Concepts:** Classification, Regression, Clustering, DNN
- **Programming:** Python, SQL, MATLAB, Julia
- **Machine Learning:** Scikit-learn, Pandas, NumPy, Librosa, Matplotlib, TensorFlow
- **Databases:** PostgreSQL, SQLAlchemy
- **Cloud & Deployment:** Docker, AWS EC2, Amazon ECR, Flask
- **DevOps:** Git, GitHub, GitHub Actions, CI/CD, PyTest
- **Tools:** VS Code, Jupyter Notebooks, QXDM, APEX, PyCharm, Wireshark,
- **Leadership & Professional Skills:** Team Coordination, Cross-functional Collaboration, Team Management, Time Management

RESEARCH EXPERIENCE

Comparison of Vocal Tract Shape Estimation Based on Autocorrelation, Covariance, Lattice and Formant Frequencies. Aug 2013 – Dec 2014

- Master's Thesis on 'Comparison of Vocal Tract Shape Estimation (VTSE) Based on Autocorrelation, Covariance, Lattice and Formant Frequencies', with Dr. Shah.
- Recorded speech signals using Praat software for male and female candidates of different age groups.
- Implemented various Linear Predictive Coding (LPC) techniques to estimate Vocal Tract Shape using MATLAB.
- Validated the obtained results with the data available from the Magnetic Resonance Imaging (MRI) technique.
- Published a research paper as a main author at ICNTE, India, (<http://ieeexplore.ieee.org/document/7029934/>).

Multi-Kernel Recursive Least Squares in the Presence of Sparse Outliers, with Dr. Slavakis, USA Aug 2018-Aug 2019

- Master's Thesis on 'Multi-Kernel Recursive Least Squares in the Presence of Sparse Outliers'. Under the Guidance of: Dr. Konstantinos Slavakis, Associate Professor, University at Buffalo.
- Successfully predicted unknown system parameters in the presence of sparse outliers with a novel optimization algorithm, titled as "Outlier Robust Multi-Kernel Recursive Least Square (OR-MKRLS) algorithm".
- Applied parallel computing in Julia to improve the performance of all the implemented algorithms, on clusters available at Center for Computational Research.

Hierarchical Clustering in Brain Network, with Dr. Slavakis, USA Jan 2018-May 2018

- Researched Hierarchical Clustering and its applications in Brain Networks.

PUBLICATIONS

Comparison of Vocal Tract Shape Estimation Based on Autocorrelation, Covariance, Lattice and Formant Frequencies. Dec 2014

- Utilized various Linear Predictive Coding (LPC) techniques to estimate Vocal Tract Shape using MATLAB.
- Published a research paper as a main author at ICNTE, India, (<http://ieeexplore.ieee.org/document/7029934/>).

CERTIFICATION

- **Real-World Product Management Course (In progress)**
- **Completed "Neural Networks and Deep Learning" course authorized by DeepLearning.AI, from Coursera. March 2025**
 - Implemented Image recognition using deep Neural Networks (NN) (Python, Numpy)
 - Trained Logistic regression, single layer NN and deep NN on 'Cat vs non-Cat' dataset with 12288 features. Deep neural network performance was best as compared to single layer neural network and logistic regression.

- Completed “Natural Language Processing in TensorFlow” course authorized by DeepLearning.AI, from Coursera.

HONORS & AWARDS

“Smart Chef” Issued by Blackstone LaunchPad, University at Buffalo

Dec 2018

- Won 2nd place at ‘Inclusive Launch Foundry Program’ for “Smart Chef Mobile Application” at ‘University at Buffalo’.
- Entrepreneurship Education.

Gold Medalist Issued by Dr. Babasaheb Ambedkar Technological University. Jan 2009

- Gold Medalist in Volleyball at Dr. Babasaheb Ambedkar Technological University.

EXTRA-CURRICULAR ACTIVITIES

- Grader for “EE531: Probability and Stochastic Processes course”.
- Student Assistant in “Campus Dining & Shops”